

• 6.041 - Probabilistic Systems Analysis - Problem Set 1

• Problem Set 1

1. a) ~~$P(A \cap A \cup B \cup C)$~~

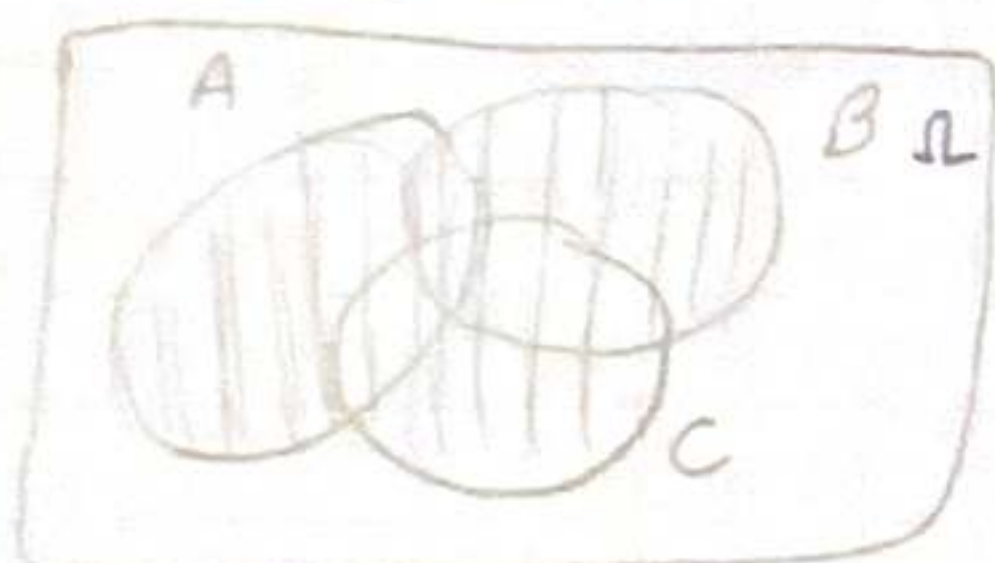
b)

c) ~~$P(A \cup B)$~~ ~~$1 - P(A \cup B \cup C)$~~ $1 - (A \cup B \cup C)$

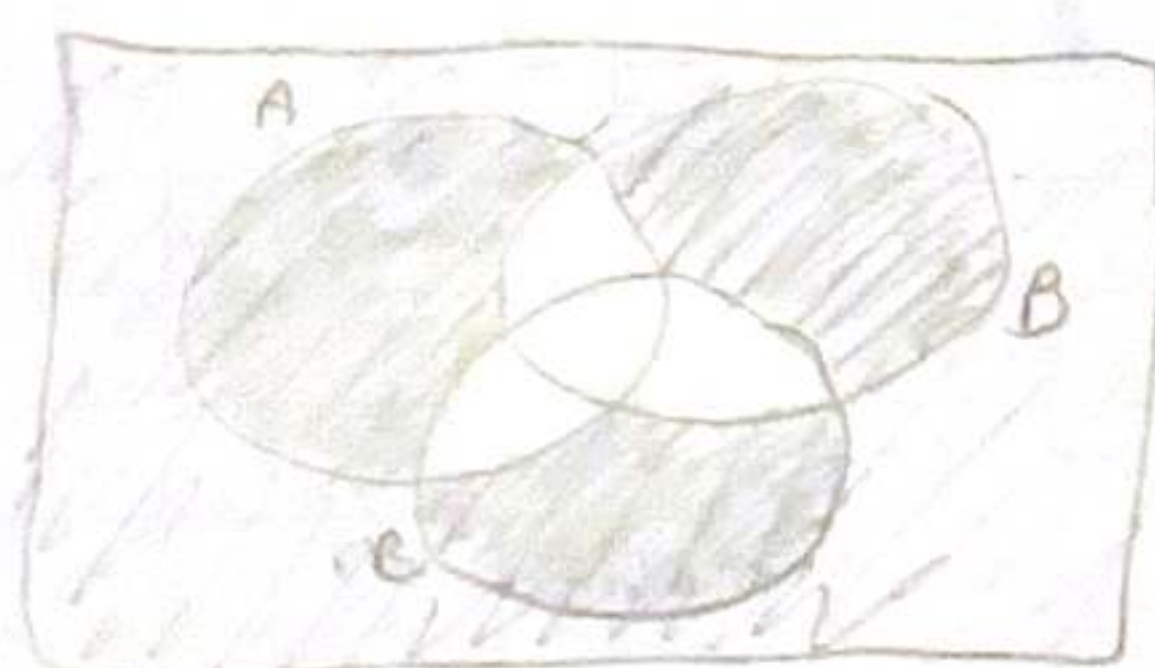
d) ~~$A \cap B \cap C$~~

e)

a) ~~$A \cap A \cup B \cup C$~~ ✓



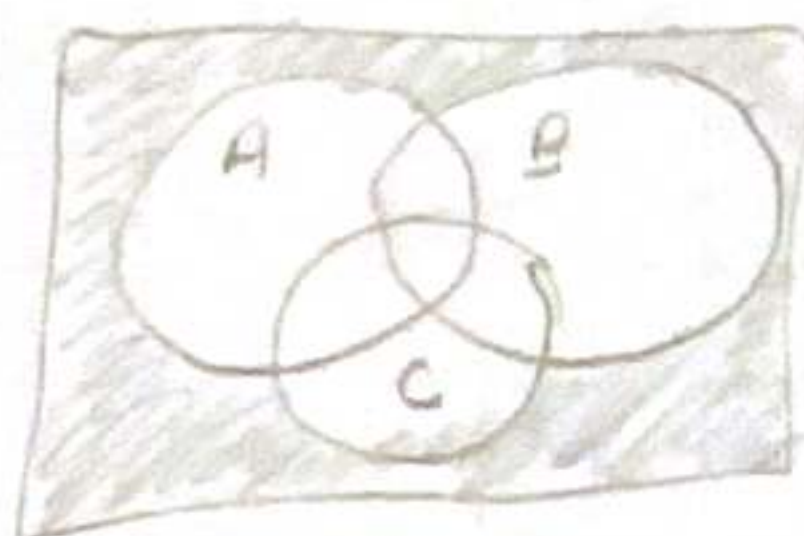
b) ~~$(A \cup B^c \cup C^c) \cup (A^c \cup B \cup C^c) \cup (A^c \cup B^c \cup C)$~~



$\cup (A^c \cap B^c \cap C^c)$

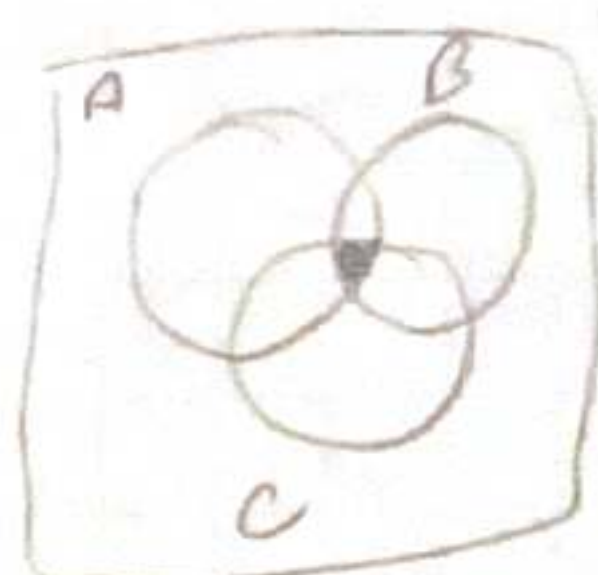
A or B or C or everything else but not (A and B)(A and C)(B and C) and not (A and B and C)

c) ~~$A^c \cup B^c \cup C^c$~~ $(A^c \cap B^c \cap C^c)$ ✓

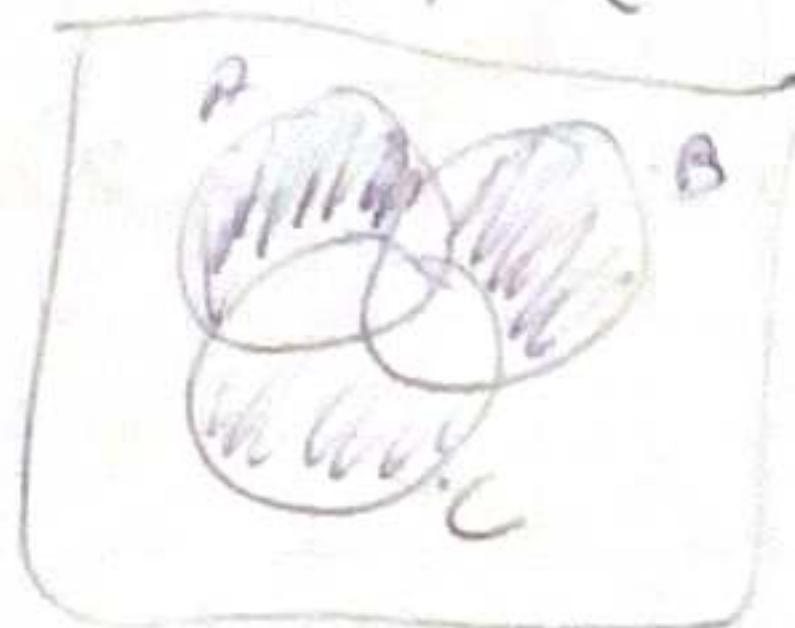


$\equiv (A \cap B \cap C)^c$

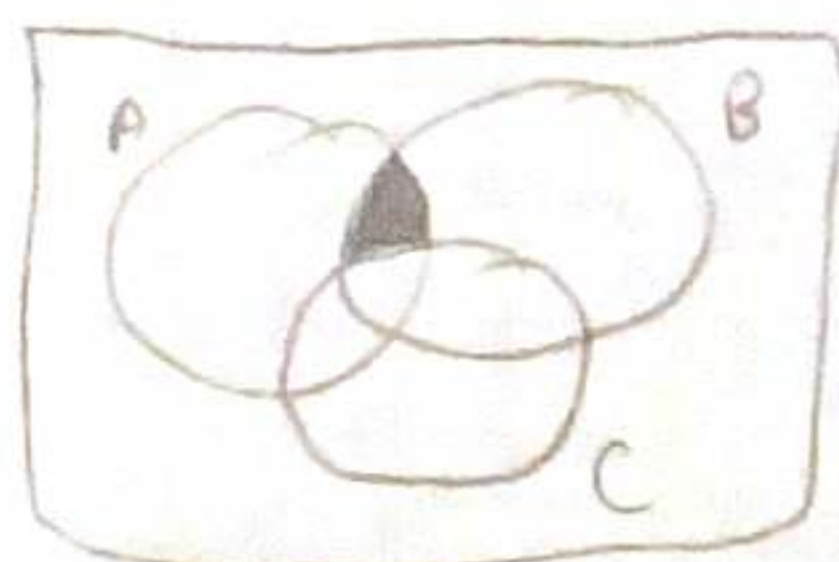
d) $(A \cap B \cap C)$ ✓



e) $(A \cup B^c \cup C^c) \cup (A^c \cup B \cup C^c) \cup (A^c \cup B^c \cup C)$



f) ~~$A \cup B \cup C^c$~~
 $A \cap B \cap C^c$ ✓



g) $A \cup B | A^c$ x
 $A \cup (A \cap B)^c$



2) Flip coin 3x.
Uniform distribution.

a) ~~HHH~~ $P(\text{HHH}) = 3 \cdot \frac{1}{2} = \frac{3}{2}$

b) ~~$P(\text{HTH}) = 3 \cdot \frac{1}{2} = \frac{3}{2}$~~

c) ~~$P(\text{HHT}) + P(\text{THH}) + P(\text{HTH}) = 3 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = \frac{9}{2}$~~

d) ~~$P(2\text{H}1\text{T}) + P(3\text{H}0\text{T}) = \frac{9}{2} + \frac{3}{2} = \frac{12}{2} = 6$~~

a) ~~$P(\text{HHHH})$~~

$P(3\text{H}0\text{T}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ ✓

b) $P(\text{HTH}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ ✓

c) $P(2\text{H}1\text{T}) = P(\text{HHT}) + P(\text{HTH}) + P(\text{THH})$
 $= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3$
 $= \frac{3}{8}$ ✓

d) $P(2\text{H}1\text{T}) + P(3\text{H}0\text{T}) = \frac{3}{8} + \frac{1}{8}$
 $= \frac{4}{8}$
 $= 0.5$ ✓

3) 4-sided die (2x).

a) $\begin{matrix} 1 & 2 & 3 & 4 & 1 & 1 & 1 & 4 \\ 1 & 2 & 3 & 4 & 2 & 3 & 4 & 7 \end{matrix}$

$\begin{matrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 3 & 4 \end{matrix}$

$2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8$

$P(\text{even sum}) = \frac{4}{8} = \frac{1}{2}$ ✓

b) $P((2;3) \cup (3;2)) = P(2;3) \cup P(3;2)$
 $= \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$
 $= \frac{2}{16} = \frac{1}{8}$ ✓

$P(\text{All outcomes}) = 80p$
 $p = \frac{1}{80}$

a) $P(\text{even sum}) = 2p + 3(4p) + 3(6p) + 1(8p)$
 $= \frac{2}{80} + \frac{12}{80} + \frac{18}{80} + \frac{8}{80} = \frac{40}{80} = \frac{1}{2}$

$P((2;3) \cup (3;2)) = 5p + 5p = 5\left(\frac{1}{80}\right) + 5\left(\frac{1}{80}\right) = \frac{10}{80} = \frac{1}{8}$

1	1	2	1
1	2	3	-
1	3	4	-
1	4	5	-
2		3	-
.		4	-
.		5	-
.		6	-
3	1	4	-
.		5	-
.		6	-
.		7	-
4	1	5	-
.		6	-
.		7	-
4		8	-

4.) A → choose Random [0; 2]
B → choose

~~P(B) = ?~~

~~P(C) = ?~~

~~P(A ∩ D) = ?~~

~~P(B) = P(Ω) - P(at least one)~~
~~= 1 - (1/3)^2 = 8/9~~

~~P(C) = 0~~
~~= 1/2 * (2) * (2)~~
~~= 1/2~~

~~P(A ∩ D) = |x - y| = 1/3~~

~~abs(x - y) = -1/3 ≤ x - y ≤ 1/3~~

~~0 ≤ x - y ≤ 1/3 ; y ≥ 1/3~~

~~y ≤ x ≤ y + 1/3~~

~~Area = (2 - 1/3) / (5/3) = 1~~

~~1/3 - 0 = 1/3~~
~~2 - 5/3 = 1/3~~

y	1/3	→	2
x	0	→	5/3
x	2/3	→	2
y	1/3	→	5/3

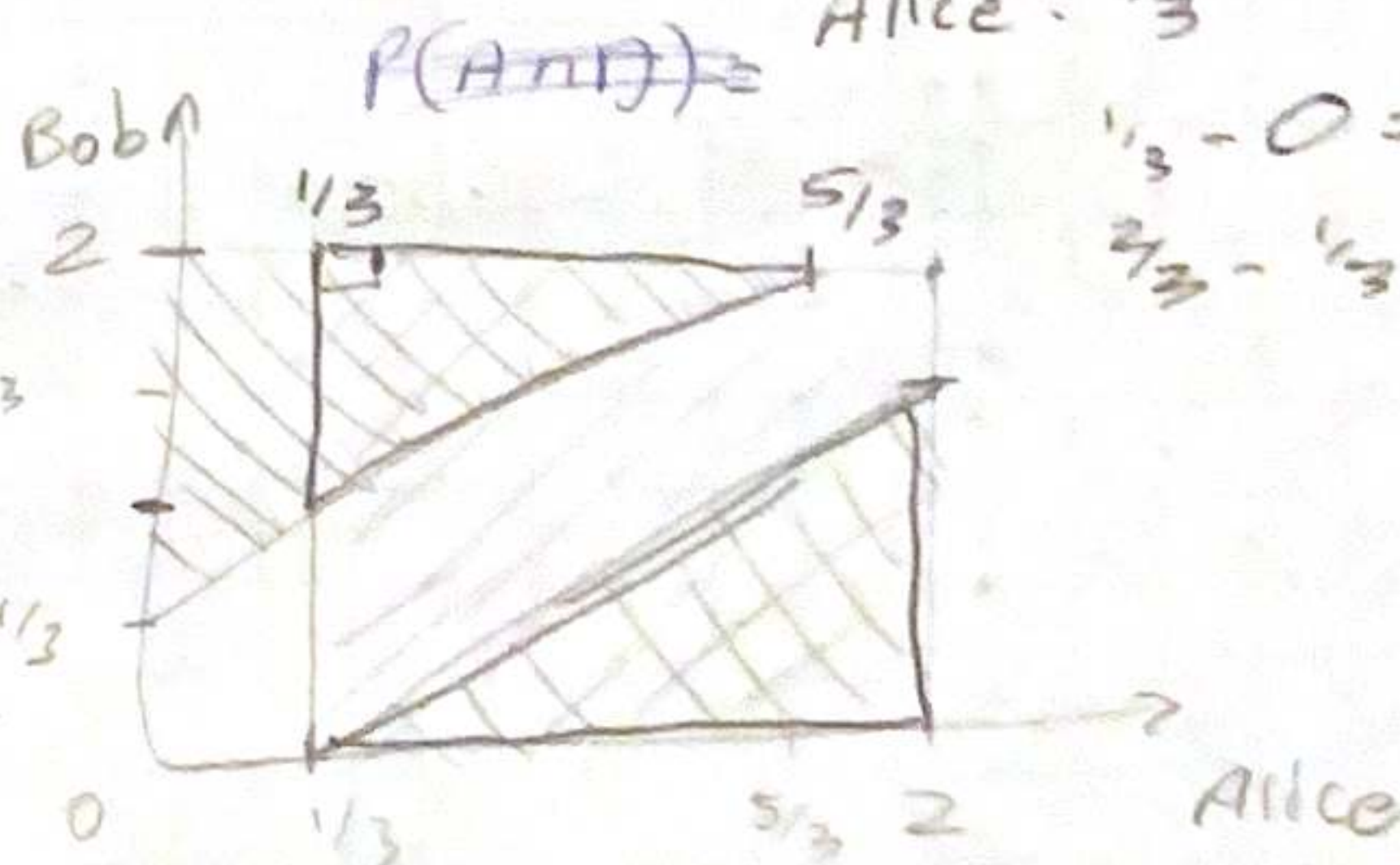
~~P(A ∩ D) = (5/3)^2 + (1/3)(4/3)~~
~~= 25/9 + 4/9~~
~~= 29/9~~

~~= 1 - (2)(1/3) = 1/3~~
~~= 1/3 - (1/3)^2~~
~~= 1/3 - 1/9~~
~~= 2/9~~
~~= 0.222~~

~~P(B) = 1 - (1/3)^2 / (2)^2~~
~~= 35/36~~

~~P(C) = area of line / total sample area~~
~~= 0~~

~~P(A ∩ D) = 1/2(2 - 1/3)(5/3) + 1/2(4/3)~~
~~= double shaded area / sample area~~
~~= 1/2(5/3)(5/3) + 1/2(4/3)(2 - 2/3)~~
~~= 25/18 + 16/18~~
~~= 41/18 / 4 = 41/72~~



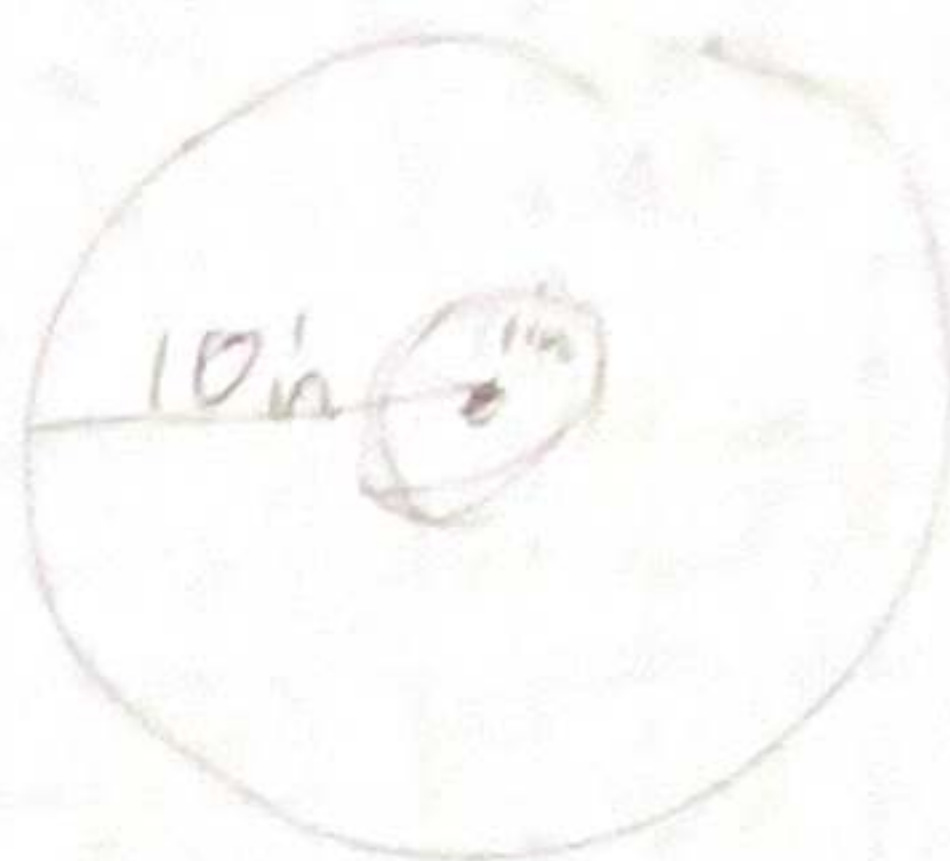
Bob: 0 or ≥ 2/3
Alice: 1/3

1/3 - 0 = 1/3
2/3 - 1/3 = 1/3

5) Mike & John play darts

Board $\Rightarrow r = 10\text{in}$

Radial [in]	Points
0-1	50
1-3	30
3-5	20
5-10	10



a) $P(\text{Mike score 50 pt 1 throw})$

$$= \frac{\text{area 50pt region}}{\text{area board}}$$

$$= \frac{\pi(1)^2}{\pi(10)^2} = \frac{1}{100} \checkmark = 0.01$$

b) $P(\text{Mike score 30pt 1 throw})$

$$= \frac{\text{area 30pt+50pt region} - \text{area 50pt region}}{\text{area board}}$$

$$= \frac{\pi r^2 - \pi r^2}{\pi r^2} = \frac{3^2 - 1^2}{10^2}$$

$$= \frac{8}{100} = 0.08 \checkmark$$

c) $P(\text{John})$

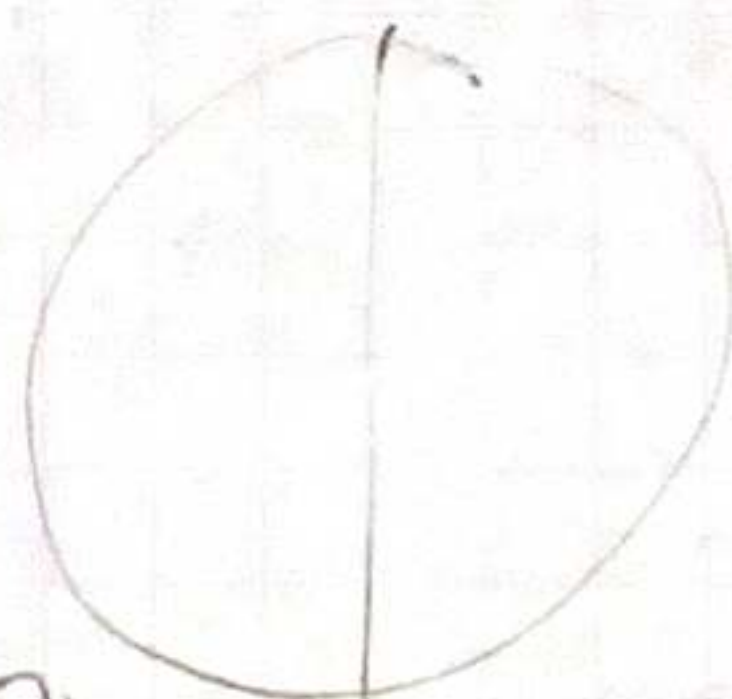
i) $P(\text{John score 50pt})$

$$= \frac{1}{3} \left(\frac{1}{2} \frac{\text{area 50pt region}}{\text{area board}} \right) + \frac{2}{3} \left(\frac{1}{2} \frac{\text{area 50pt region}}{\text{area board}} \right)$$

$$= \frac{1}{3} \left(\frac{1}{2} \frac{(1)^2}{100} \right) + \frac{2}{3} \left(\frac{1}{2} \frac{1}{100} \right)$$

$$= \frac{1}{600} + \frac{2}{600} = \frac{3}{600}$$

$$= \frac{1}{200} \checkmark$$



$\rightarrow 2 \times$ likely

RHS 50pt + LHS 50pt

$$P(\text{RHS 50pt}) = \alpha \frac{1}{2} \pi R^2 \quad (r=1)$$

$$= \frac{\alpha \pi}{2}$$

$$P(\text{LHS 50pt}) = \beta \frac{1}{2} \pi R^2 \quad (r=1)$$

$$= \frac{\beta \pi}{2}$$

$$P(\text{RHS}) = \alpha \pi R^2 / 2 \quad (r=10)$$

$$= \alpha 50 \pi = \frac{2}{3}$$

$$\alpha = \frac{2}{75 \pi}$$

ii) $P(\text{John score 30pt}) = \frac{\text{Right half 30pt reg area} + \text{Left half 30pt reg} - \text{area 50pt}}{\text{board area}}$

$$= \frac{2}{3} \left(\frac{1}{2} \frac{\pi(3)^2}{\pi(10)^2} \right) + \frac{1}{3} \left(\frac{1}{2} \frac{\pi(3)^2}{\pi(10)^2} \right) - \frac{1}{200}$$

$$= \frac{18}{600} + \frac{9}{600} - \frac{1}{200}$$

$$= \frac{9}{600} - \frac{1}{200} = \frac{3}{200} - \frac{1}{200}$$

$$= \frac{2}{200}$$

$$P(\text{LHS}) = \frac{\beta \pi}{2}$$

$$\frac{1}{3} = \frac{\beta \pi}{2}$$

$$\beta = \frac{1}{150 \pi}$$

$$P(\text{RHS 50pt} + \text{LHS 50pt}) = \left(\frac{2}{75 \pi} \right) \left(\frac{\pi}{2} \right) + \left(\frac{1}{150 \pi} \right) \left(\frac{\pi}{2} \right)$$

$$= \frac{1}{150} + \frac{1}{300}$$

$$= 0.01$$

$$6). P(A \cap B \cap C) \geq P(A) + P(B) + P(C) - 2$$

$$P(A \cap B \cap C) = P((A \cap B) \cap C)$$

$$= P(A \cap B) + P(C)$$

$$= P(A) + P(B) - P(A \cup B) + P(C)$$

$$P(A \cap B) + P(C) = P(A) + P(B) + P(C) - P(A \cup B)$$

$$P(A \cap B \cap C) = P((A \cap B) \cap C)$$

$$= P(A \cap B) + P(C)$$

$$=$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

ii) Reattempt:

$$P(\text{John scores } 30_{pt}) = P(\text{John score } 30_{pt}) - P(\text{John score } 50_{pt})$$

$$= P(RH30_{pt}) + P(LH30_{pt}) - P(50_{pt})$$

$$P(RH30_{pt}) = \frac{1}{2} \alpha \pi r^2 = \frac{\alpha \pi r^2}{2} \quad (r=3)$$

$$= \frac{\alpha 9\pi}{2} \quad \text{Need } \alpha$$

$$P(RHB) = \frac{2}{3} = \frac{\alpha \pi r^2}{2} \quad (r=10)$$

$$\frac{2}{3} = \frac{\alpha 100\pi}{2} = \alpha 50\pi$$

$$\alpha = \frac{1}{75\pi}$$

$$P(RH30_{pt}) = \left(\frac{1}{75\pi}\right) \left(\frac{1}{2}\right) (\pi^2) = \frac{1}{150}$$

$$P(LH30_{pt}) = \frac{1}{3} = \frac{\beta \pi (100)}{2}$$

$$\frac{1}{3} = \frac{\beta \pi 100}{2}$$

$$P(LH30_{pt}) = \frac{1}{2} \beta \pi r^2 \quad (r=3)$$

$$= \frac{9}{2} \beta \pi$$

Need β

$$P(LHB) = \frac{1}{3} = \frac{\beta \pi r^2}{2} \quad (r=10)$$

$$\frac{1}{3} = \frac{\beta \pi 100}{2}$$

$$\frac{1}{3} = \beta \pi 50$$

$$\beta = \frac{1}{150\pi}$$

$$\frac{1}{3} = \frac{1}{2} \frac{\beta \pi (10)^2}{1}$$

$$\frac{2}{3} = \beta \pi 100$$

$$\beta = \frac{2}{300\pi} = \frac{1}{150\pi}$$

$$\therefore P(LH30_{pt}) = \frac{9}{2} \left(\frac{1}{150\pi}\right) \pi$$

$$= \frac{9}{2} \times \frac{1}{150} = \frac{3}{100}$$

$$= \frac{1}{2} \beta \pi r^2$$

$$= \frac{1}{2} \left(\frac{1}{150\pi}\right) (3-1)^2 \pi$$

$$= \frac{2}{150} = \frac{1}{75}$$

$$\frac{2}{3} = \frac{1}{2} \alpha \pi r^2 \quad (r=10)$$

$$\frac{2}{3} = \frac{1}{2} \left(\frac{1}{75\pi}\right) \pi (3-1)^2$$

$$\frac{2}{3} = \frac{2}{75}$$

$$= \frac{2}{150} + \frac{2}{75}$$

• Introduction to Probability

↳ ~~Text~~ Textbook Problems pg 54

27-04-2026

• Section 1.2 Problems Continued

$$1.2.10) P((A \cap B^c) \cup (A^c \cap B)) = P(A) + P(B) - 2P(A \cap B)$$

$$\Rightarrow P(A \cap B^c) = P(A) - P(A \cap B)$$

$$\Rightarrow P((P(A) - P(A \cap B)) \cup (P(A^c) - P(A^c \cap B)))$$

$$\Rightarrow P([P(A) + P(A^c)] - [P(A \cap B) + P(A^c \cap B)])$$

$$P(A \cap B)$$

$$\Rightarrow P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A) + P(B) - 2P(A \cap B)$$

$$\Rightarrow P((A \cap B^c) \cup (A^c \cap B))$$

$$= P(A \cap B^c) + P(A^c \cap B)$$

$$= P(A) - P(A \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cap B^c) + P(A^c \cap B)$$

$$\Rightarrow P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A) + P(B) - 2P(A \cap B)$$

$$\Rightarrow P(A) + P(B) - 2P(A \cap B)$$



1.2.12) a)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \cancel{P(A \cup B \cup C)} = \cancel{P(A \cup B)}$$

$$\Rightarrow P(A \cup B \cup C) = P(A \cup B) + P(C) - P((A \cup B) \cap C)$$

$$= P(A \cup B) + P(C) - P((A \cap C) \cup (B \cap C))$$

$$= P(C) + P(A \cup B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

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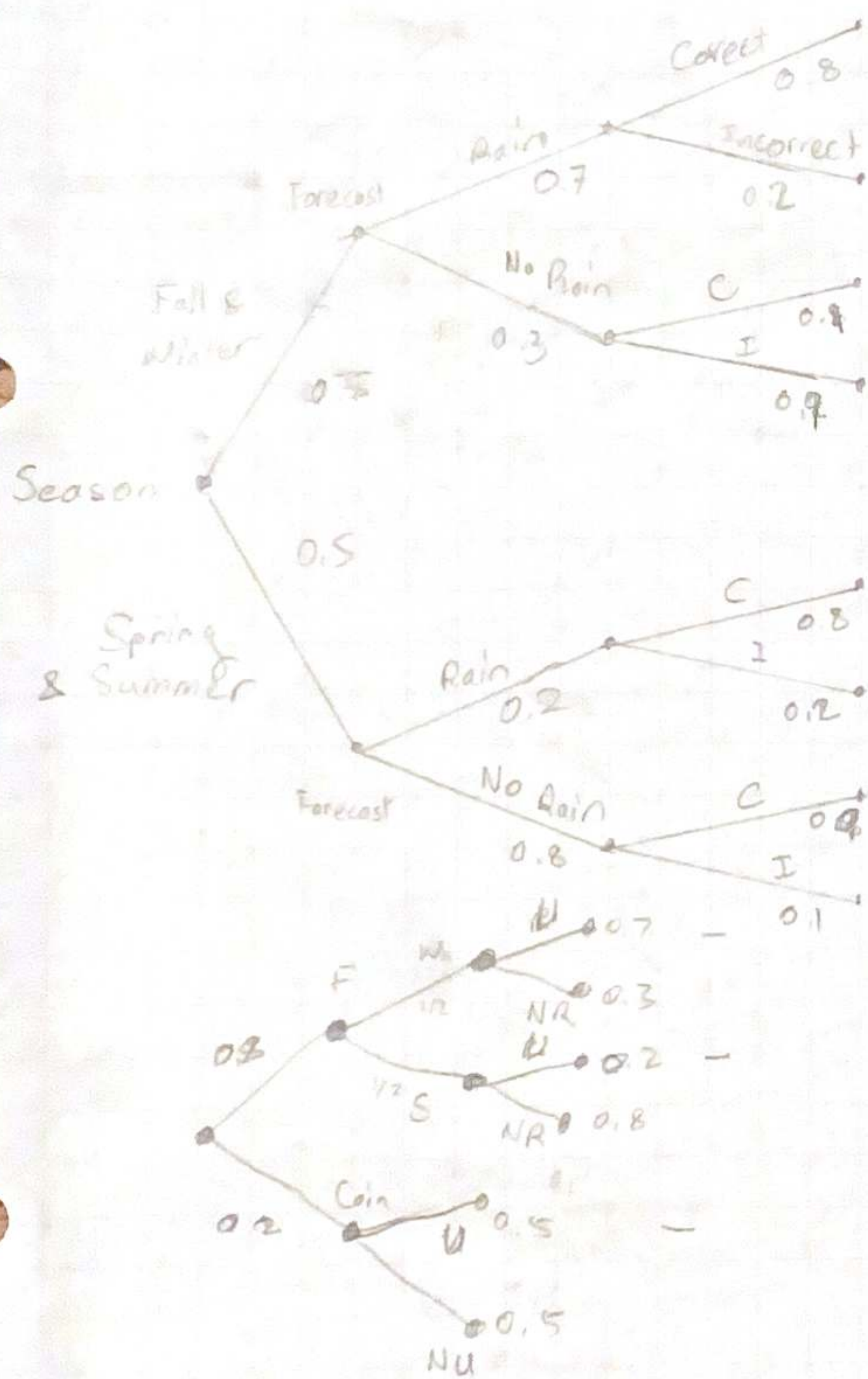
Introduction to Probability (6.041 Probabilistic Systems Analysis)

23-04-2026.

Problem Set 2

1a) Forecast $\rightarrow$ Rain $\rightarrow P(\text{Rain}) = 0.8$
 No Rain $\rightarrow P(\text{Rain}) = 0.1$

Fall/Winter Forecast $\rightarrow$ Rain $\rightarrow P(\text{Forecast Rain}) = 0.7$
 Summer/Spring Forecast $\rightarrow$ Rain $\rightarrow P(\text{Forecast Rain}) = 0.2$



a) $P(\text{Rain} | \text{Winter}) = 0.7(0.5) = 0.35$
 $P(\text{Rain} | \text{Summer}) = 0.2(0.5) = 0.1$

b) Victor miss forecast = 0.2
 Given miss, flip coin = 0.5

$$P(A \cap B) = P(A \cap B)$$

$$P(A \cap B | C) = P(A | C) \cdot P(B | C)$$

U = carry umbrella
 K = forecast no rain
 J = miss forecast
 $P(U \cap K)$

$$P(U) = (0.2)(0.5) + (0.8)(0.5)(0.2) + (0.8)(0.5)(0.7)$$

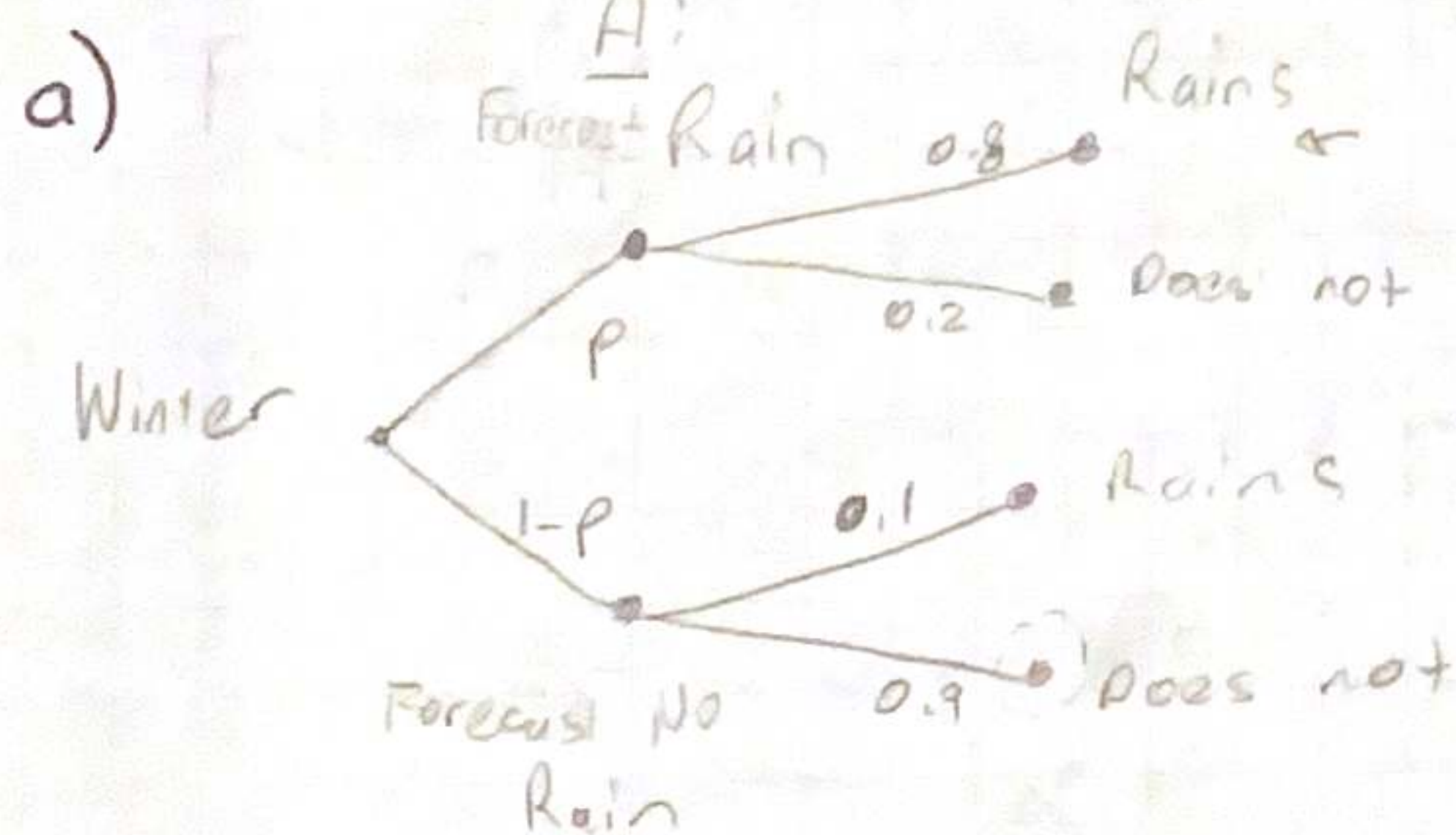
$$= 0.46$$

$$P(N) = (0.5)(0.3) + (0.5)(0.8)$$

$$= 0.55$$

$$P(U | P(N)) = (0.46)(0.55) = 0.253$$

$$P(U \cap N) = 1$$



$p = 0.7$ in winter
 $p = 0.2$

Given: Winter; actually rained

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

$$= \frac{(0.8)(0.7)}{(0.7)(0.8) + (1-0.7)(0.1)}$$

$$= \frac{56}{59} = 0.94915$$

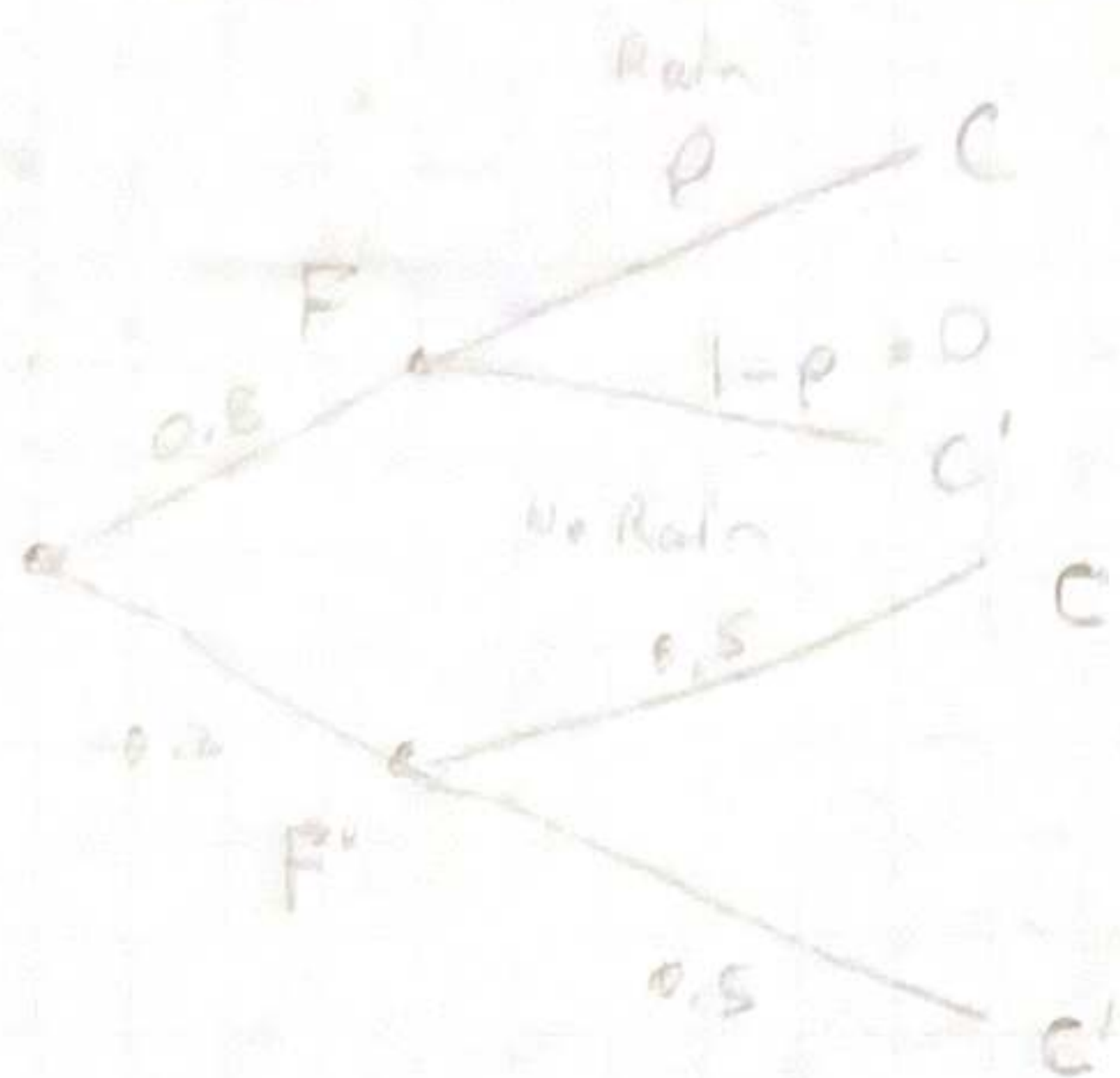
$$= 94.92\%$$

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

$$= \frac{(0.2)(0.8)}{(0.2)(0.8) + (1-0.2)(0.1)}$$

$$= \frac{2}{3}$$

b)



F - Forecast
F' - missed forecast
C - carrying umbrella
C' - no umbrella.

C - Victor carrying an umbrella
N = (C'/F) - the forecast is no rain = 0

$$P(C) = (0.2)(0.5) + (0.8)(p) = 0.8p + 0.1$$

$$P(D) = 1 - p$$

$$P(C/D) = (0.2)(0.5) + (0.8)(1-p)$$

$$P(D) = (1-p) = 0$$

(However give D → no umbrella
so that event cannot be
> 0.)

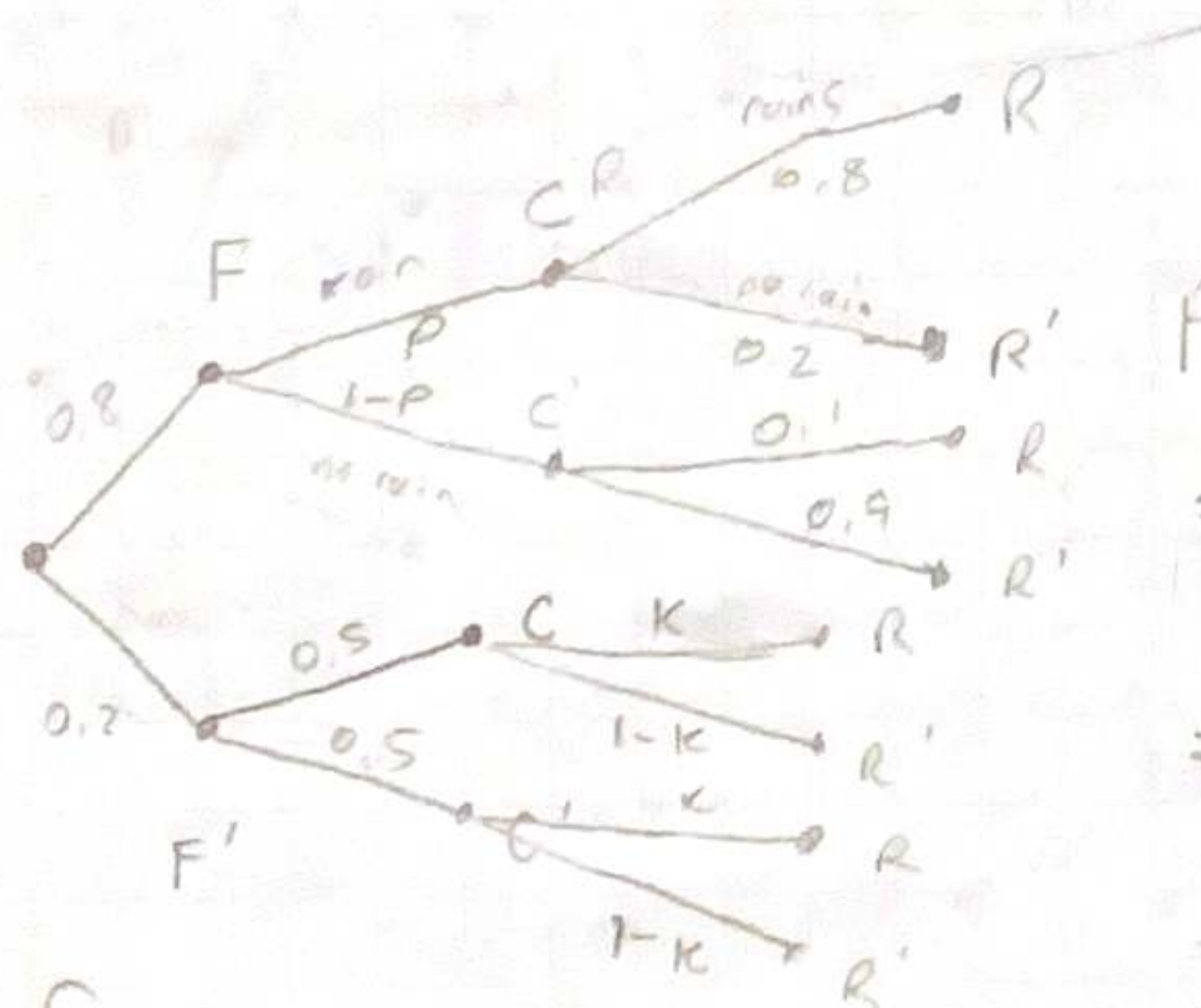
$$P(C/D) = 0.1$$

∴ $P(C) = P(C/D)$ if and only if $p = 0$

However $p = \{0.7, 0.2\}$

∴ C and D can never be independent, and the result does not depend on the season.

c)



Victor carrying Umbrella and not raining

$$P(\text{saw forecast} | \text{Umbrella + not raining}) = \frac{P(F \cap (C \cap R'))}{P(C \cap R')}$$

$$= \frac{P(F) P((C \cap R') | F)}{P(C \cap R')}$$

$$= \frac{(0.8)(p)(0.2)}{(0.8)(p)(0.2) + (0.2)(0.5)(1-k)}$$

$$P(k) = P(\text{rains} | \text{missed forecast}) =$$

F - saw forecast
F' - missed forecast
C - umbrella
C' - no umbrella
R - actually rains
R' - no rains

- Introduction to Probability - Textbook Problems

- Problems - Section 1.1 - Sets

1) Roll 6-sided die

	1	2	3	4	5	6
A - even number						
B - roll greater than 3						

$$(A \cup B)^c = A^c \cap B^c \quad \text{O} \quad (A \cup B)^c = \{1, 3\}$$

$$A^c \cap B^c = 1 - \frac{2}{6} = \frac{4}{6}$$

$$A^c \cap B^c = \{1, 3, 5\} \cap \{1, 2, 3\}$$

$$= \{1, 3\}$$

$$\therefore (A \cup B)^c = A^c \cap B^c$$

③ $(A \cap B)^c = (\{1, 4, 6\})^c$ $A^c \cup B^c = \{2, 4, 6\}^c \cup \{4, 5, 6\}^c$
 $= \{1, 2, 3, 5\}$ $= \{1, 3, 5\} \cup \{1, 2, 3\}$

$$= \{1, 3, 5\} \cup \{1, 2, 3\}$$

$$= \{1, 2, 3, 5\}$$

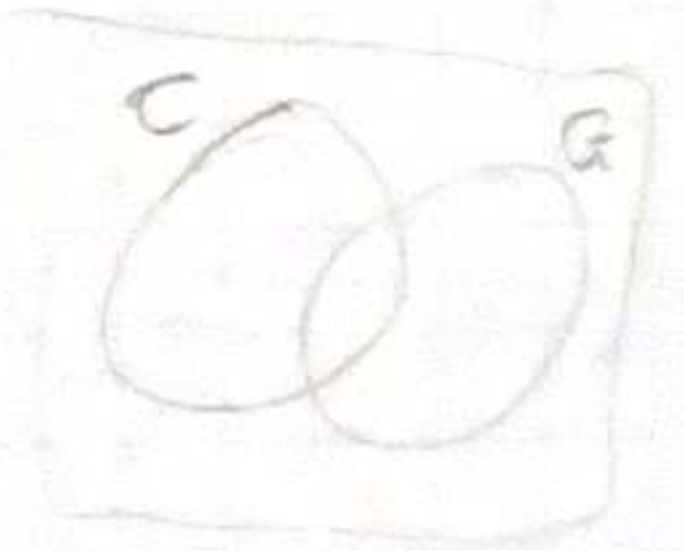
$$\therefore (A \cap B)^c = A^c \cup B^c$$

~~2) a) Prove $A^c = (A^c \cap B) \cup (A^c \cap B^c)$~~

- Problems - Section 1.2 - Probabilistic Models

1.25) $P(G) = 0.6$ $P(G \cap C) = 0.4$
 $P(C) = 0.7$

$$\begin{aligned} P(G' \cap C') &= P(\Omega) - P(G) - P(C) + P(G \cap C) \\ &= 1 - 0.6 - 0.7 + 0.4 \\ &= 1 - 0.9 \\ &= \underline{0.1} \end{aligned}$$



1.2.6) Six sided die. Unfair $\rightarrow$ even face 2x likely as odd

$$P(E) = 2/9$$

$$P(0) = 1/9$$

$$P(\text{Less than 4}) = P(\{1, 2, 3\}) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9} = 0.444$$

1.2.7) Four-sided die.
Roll repeatedly.

$$\text{Outcomes} = \{1, 2, 3, 4\}$$

Sample Space?

Outcome can be any finite sequence of form $(a_1, a_2, \dots, a_n)$,
where n is an arbitrary positive integer, $a_1, a_2, \dots, a_{n-1}$ belong to
 $\{1, 3\}$ and a_n belongs to $\{2, 4\}$

Additionally,

Section 1.3 Problems

14) a) 2x 6 sided dice $\Rightarrow \begin{bmatrix} (1,1); (1,2); \dots; (1,6) \\ (2,1); (2,2); \dots; (2,6) \\ \vdots \\ (6,1); (6,2); \dots; (6,6) \end{bmatrix}$ 36 events

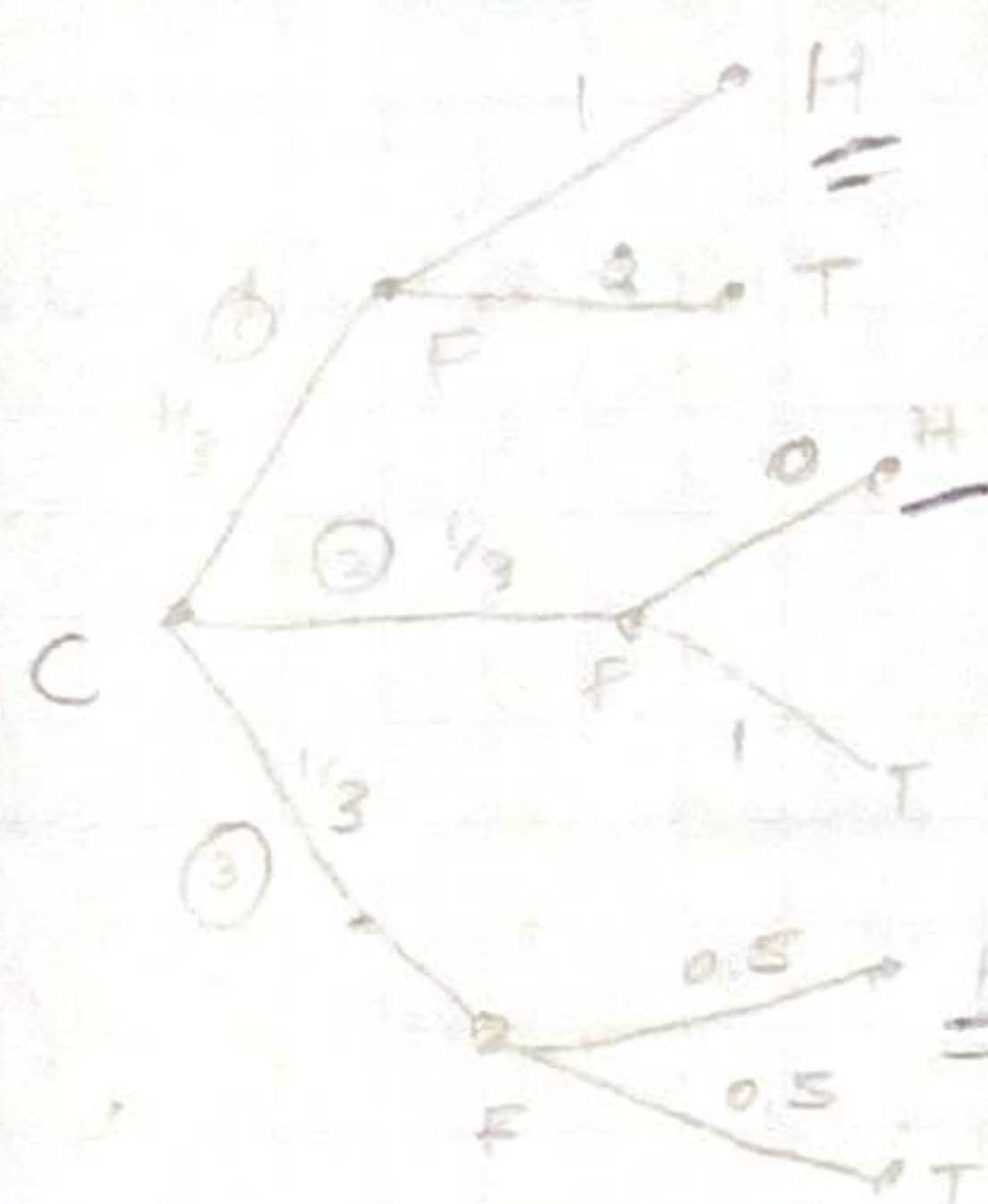
$$P(\text{doubles}) = \frac{6}{36} = \frac{1}{6} \checkmark$$

b) $P(\text{doubles} \mid \text{sum} \leq 4) = \frac{2}{36} = \frac{1}{18} \times \begin{bmatrix} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (3,1) \end{bmatrix} \Rightarrow \frac{2}{6} = \frac{1}{3}$

c) $P(\text{at least one 6}) = \frac{6+5}{36} = \frac{11}{36} \checkmark$

d) $P(\text{at least one 6} \mid \text{different numbers}) = \frac{5+5}{36} = \frac{1}{3} \checkmark$

16) 3 coins $[H, H]_1, [T, T]_2, [H, T]_3$



$P(\text{Tother face} \mid \text{heads})$

$$= \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(0.5)$$

$$= \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(0.5)$$

$$= \frac{0 + 0.5}{2}$$

$$= \frac{1}{4} \times$$

$p = P(\text{two-headed coin was chosen} \mid \text{heads came up})$

$$p = \frac{P(\text{two-headed coin was chosen and heads came up})}{P(\text{heads came up})}$$

$$= \frac{P(2HC \cap H)}{P(H)} = \frac{\frac{1}{3}}{\frac{1}{2}}$$

$$P(H) = \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(0.5) = \frac{1}{2}$$

$$= \frac{2}{3}$$

prob other face tails $= 1 - p = 1 - \frac{2}{3} = \frac{1}{3}$

17) 100 items; test 4 items randomly; $\geq \frac{1}{4}$ def time $\Rightarrow$ Reject batch.

$$P(\text{Accept} \mid \text{5 Defective}) = \left(\frac{5}{100}\right)\left(\frac{5}{99}\right)\left(\frac{5}{98}\right)\left(\frac{5}{97}\right)$$

$$\begin{aligned} P(D) &= 0.2031 \\ P(A \mid D) &= 1 - P(D) \\ &= 1 - 0.2031 \\ &= 0.7969 \end{aligned}$$

$$P(A) = \frac{95}{100} \cdot \frac{94}{99} \cdot \frac{93}{98} \cdot \frac{92}{97} = 0.812$$

$$P(A) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2)P(A_4 \mid A_1 \cap A_2 \cap A_3)$$

• Section 1.4 Problems

9) Drawer j $p_j > 0 \rightarrow$ Knows
 Drawer i $p_i = d_i \rightarrow$ Guess

$$P(\text{drawer } j) = \sum_i P(\text{Drawer } i) \cdot P(\text{Drawer } j | \text{Drawer } i)$$

$$P(\text{Drawer } j) = (p_j)(1 - \sum_i d_i)$$

Let A be event Alice does not find her paper in Drawer i

$$\therefore P(A) = 1 - \sum_i d_i$$

Let B be event paper is in Drawer j $P(A|B) = 1 - d_i$

assume $i \neq j$ $P(B) = p_j$
 ~~$P(A \cap B) = P(B)$~~

$$P(B|A) = P(B), \text{ know } A \text{ did not occur}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)}{P(A)} = \frac{p_j}{1 - \sum_i d_i} \quad \text{if } i \neq j$$

assume $i = j$

~~$P(B)$~~

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)P(A|B)}{P(A)}$$

$$= \frac{P(B) \cdot P(A|B)}{P(A)} = \frac{(p_j)(1 - d_i)}{1 - \sum_i d_i} \quad \text{if } i = j$$

~~$P(A|B) = \frac{p_j - d_i}{1 - \sum_i d_i}$~~

$$P(A|B) = 1 - d_i$$

$\hookrightarrow$ ~~if~~ $i = j$ (assume drawer $j = i$)

$\hookrightarrow$ $\because B$ happened $\Rightarrow$ paper is in $j = i$

$\hookrightarrow$ $\because$ Only depends on if Alice finds it? $d_i \Rightarrow$ finds it.

$$\therefore 1 - d_i = P(A|B)$$

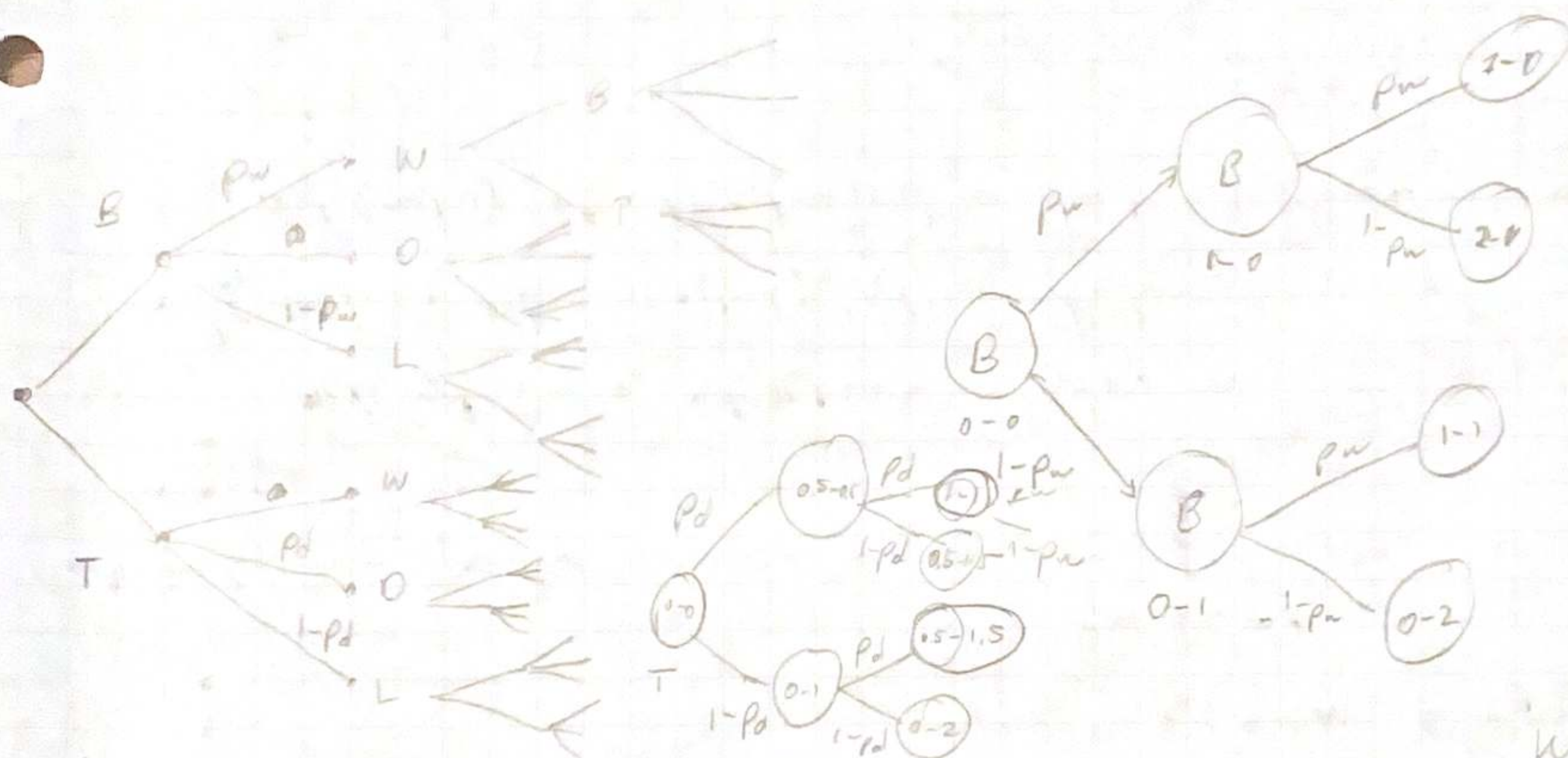
Section 1.4 → Total Probability Theorem Problems con.

20) Boris
 two styles
 → timid
 → bold.

Two game chess match
 → Each game:
 → Player 1 wins
 → Player 2 wins
 → Draw

timid play → Draw $p_d = 0$
 Lose $1 - p_w$
 win p_w

bold play → always chosen for sudden death
 → win p_w
 → Lose $1 - p_w$
 → May switch style between games 1 & 2



a) i) $P(B \text{ game } 1+2) = (p_w)(p_w) = p_w^2$

W-L-W
 L-W-W
 W-W

ii) $P(T \text{ game } 1+2) = P(\text{timid game } 1+2) + P(\text{win sudden death})$
 $= (1-p_d)(1-p_d) + p_w$
 $= (1-p_d)^2 + p_w$

i) $P(\text{Win match} | \text{Bold}) = p_w^2 + (p_w)(1-p_w)(p_w) + (1-p_w)(p_w)^2$
 $= p_w^2 + 2(p_w)^2(1-p_w)$
 $= [1 + 2(1-p_w)]p_w^2$

ii) $P(\text{Win Match} | \text{play timid}) = (p_d)(p_d)(p_w)$
 $= p_w p_d^2$

iii)

$$\begin{aligned} P(\text{Win} \mid \text{Strategy}) &= \underline{p_w p_d} + \underline{p_w(1-p_d)(p_a)} + (1-p_w)^2 p_w \\ &= p_w p_d + p_w^2(1-p_d) + p_w(1-p_w) p_a \\ &= p_d p_w + (1-p_d) p_w^2 + (1-p_w) p_a^2 \end{aligned}$$

b) $p_w < 1/2$ (less 50% win chance $\Rightarrow$ less skilled than opponent)
 $p_w = 0.45$

$$P(\text{Boris Wins}) = (\quad)(\quad)$$